



Benefits of thymol supplementation on performance, the hepatic antioxidant system, and energetic metabolism in grass carp

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Abstract Thymol is an herbal food additive used to improve animal performance. Thymol acts via its potential to enhance productive and reproductive performance, and by improving bioavailability of nutrients in fish. Nevertheless, the exact mechanisms associated with these phenomena remain poorly understood, although recent evidence has suggested the involvement

Highlight

- Thymol is an herbal food additive used to improve animal performance.
- Dose of 100 mg thymol/kg dietary supplementation tested in the present study increased the weight gain of fingerlings of grass carp under non-ideal conditions.
- Thymol dietary supplementation (100 mg/kg feed) was able to improve the hepatic energy metabolism.
- Thymol in diet enhanced the hepatic antioxidant status, which can be pathways involved in the increase of growth performance of fingerlings grass carp.

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of the phosphotransfer network and antioxidant status. Therefore, the aim of this study was to determine whether the improvement of the antioxidant/oxidant status and the phosphoryl transfer network may be involved in enhanced growth performance in grass carp (*Ctenopharyngodon idella*) fed with various levels of thymol (100, 200, and 300 mg/kg feed). Thymol-supplementation (100 mg/kg feed) produced higher body weight and weight gain for 60-day post-feeding compared to the control group. Specific growth rate was higher; while feed conversion ratio was lower in fish that consumed 100 mg of thymol/kg compared to other groups. Hepatic lactate dehydrogenase activity and lipid peroxidation levels were lower in the thymol-supplemented group (100 mg/kg feed) than in the control group, while reactive oxygen species were lower in all supplemented groups than in the control group. Hepatic superoxide dismutase (300 mg/kg feed) and glutathione peroxidase (100, 200, and 300 mg/kg feed) activities, as well as antioxidant capacity against peroxyl radicals (100 mg/kg feed) were higher in these groups than in the control group. Based on these data, we conclude that 100 mg thymol/kg dietary supplementation increased growth performance of fingerling grass carp. Finally, hepatic adenylate kinase activity was lower in the thymol supplemented group (100 mg/kg feed) than in the control group. Thymol supplementation (100 mg/kg feed) improved hepatic energy metabolism, while practically all tested concentrations of thymol enhanced hepatic antioxidant status, all of which may be pathways involved in increased growth performance in fingerling grass carp.

Keywords Phosphoryl transfer network · Aquaculture · Fish performance · Free radicals

Introduction

Aquaculture is the fastest growing food production sector in the world with an average annual growth rate of 5.8% during 2000–2016, reaching 80 million tons of food fish in 2016 (FAO 2018). This rapid growth occurred because aquaculture is responsible for the supply of fish for human consumption, and fish and fishery products represent a source of proteins and essential micronutrients for human health (Marinho et al. 2013). To supply the demand, a number of antibiotics have been used in animal diets as growth promoters in order to avoid disease, improve performance, and increase intestinal microbial population (Bento et al. 2013). The emergence of bio-resistance, the transmission of resistant microorganisms from animals to humans, as well as the increasing demand for aquaculture friendlier to the environment all have driven the search for natural growth promoters (Amer et al. 2018).

Thymol (2-isopropyl-5-methylphenol) is a natural monoterpene found in some essential oils extracted from plants, including thyme (*Thymus vulgaris*) (Hoseini and Yousefi 2019) and oregano (*Origanum heracleoticum*) (Zheng et al. 2009). Thymol has been successfully used in fish nutrition as an herbal feed additive to improve performance, to activate digestive system structure and function, to enhance metabolism, and to reduce the damage of compounds/free radicals derived from interactions with cellular biological compounds (Ran et al. 2016; El-Hack et al. 2016; Anyu et al. 2018). In fish, thymol has been shown to be an excellent anesthetic in several fish species, including the common carp (*Cyprinus carpio*) (Yousefi et al. 2018) and silver catfish (*Rhamdia quelen*) (Bianchini et al. 2017), with antibacterial activity against *Aeromonas hydrophila* (Da Cunha et al. 2019), as well as growth-promoting properties (Anyu et al. 2018). In this sense, a recent study conducted by Amer et al. (2018) showed that the 969 and 1938 mg of thymol/kg of diet improved growth performance of Nile tilapia (*Oreochromis niloticus*) fingerling over 75 days. These authors also reported that thymol improved muscle antioxidant/oxidant status via reduction of lipid peroxidation and augmentation of glutathione reductase levels, concluding that the improvement of antioxidant/oxidant status is a pathway associated with enhancement of fish

performance. Some studies have suggested that the improvement of antioxidant/oxidant status exerts beneficial effects on enzymes belonging to the phosphotransfer network, including adenylate kinase (AK) and pyruvate kinase (PK) (Baldissera et al. 2019). Both enzymes mediate energetic metabolism associated with adenosine triphosphate (ATP)-production and utilization in tissues with high and fluctuating energy requirements, including the liver (Dzeja and Terzic 2003).

The phosphoryl transfer network plays an essential role in energy transfer and distribution, and it secures the cellular economy by maintaining the balance between ATP consumption and production (Dzeja and Terzic 2003). PK is an irreversible enzyme of in glycolysis pathway that forms pyruvate and ATP, playing a fundamental role in suitable energy production in tissues with high energy demands (Wang et al. 2002). AK also maintains bioenergetic balance by reconverts two adenosine diphosphate (ADP) molecules into one ATP and one adenosine monophosphate (AMP), providing a doubling of energetic potential, a crucial step for improvement of intracellular energetic communication, and metabolic monitoring in tissues with high and fluctuating demands for energy (Dzeja and Terzic 2009). Baldissera et al. (2019) demonstrated that the reduction of hepatic lipid peroxidation and the increase of total antioxidant capacity caused an augmentation of hepatic PK activity, directly contributing to improved growth performance in Nile tilapia fed with vegetable choline (phosphatidylcholine). Therefore, our hypothesis is that the improvement of the hepatic antioxidant/oxidant elicited by thymol supplementation will enhance the phosphoryl transfer network, thereby having beneficial effects on fish performance.

In many countries where grass carp are bred, there are well-marked seasons of the year. In winter, temperatures are lower, and grass carp was acclimated to 24 °C (Galloway and Kilambi 1984). This is the period when farmed fish receive only ration as food, because of low availability of plants used commonly for feed (in herbivorous fish); during this period, growth generally stops. The use of additives, particularly thymol, may be an alternative during periods of non-ideal environmental conditions for grass carp. The aim of the present study was to determine whether improvement of antioxidant/oxidant status and phosphoryl transfer network are pathways positively associated with growth performance in grass carp (*Ctenopharyngodon idella*) fed with various levels of dietary thymol.

Materials and methods

Thymol and diet preparation

Thymol (molecular weight, 150.22 g/mol) was purchased from Sigma-Aldrich® (São Paulo, Brazil). The basal diet was based on corn, poultry viscera meal, feather meal, and soybean meal, which were crushed and mixed with other ingredients (Table 1), providing all nutritional requirements for fish (National Research Council 2011). For supplemented diets, thymol (0.0, 100, 200, and 300 mg/kg of feed) was added after preparation of the basal diet. Thymol was included in the feed after the extrusion process. The granulated thymol was powdered using a porcelain crucible and stick. Then the thymol was mixed into the basal feed for 10 min using Y blender (2 kg

Table 1 Centesimal, calculated, and analyzed compositions of fish diet without thymol: NRC (2011)

Ingredients	g/kg
Ground corn	320.0
Poultry viscera meal	280.1
Feather meal	280.0
Soybean meal ¹	74.4
Limestone	40.0
Vitamins and Minerals premix ²	4.00
Antifungal	1.00
Antioxidant	0.50
Calculated composition	
Dry matter, %	90.6
Crude protein, %	42.0
Digestible energy, Mcal/kg	3.25
Calcium, %	2.77
Non-phytate phosphorus, %	0.98
Sodium, %	0.19
Neutral detergent fiber	4.79
Acid detergent fiber	1.74

¹ Product with 46% of crude protein; Cooperalfa, Brazil; ² 1 kg of vitamins and minerals premix provided the minimum of Calcium 166 g; copper 357 mg; chrome 9.4 mg; iron 2.34 g, phosphorus 35 g; carnitine 1041.6 mg; phytase 11666 FTU; lysine 46.67 g; manganese 1.17 g; selenium 10.5 mg; sodium 40.8 g; zinc 2.81 g; folic acid 93.70 mg; pantothenic acid 468 mg; biotin 10.55 mg; choline 12.50 g; niacin 703 mg; vitamin A 233000 UI; vitamin B12 703 µg; vitamin B2 117 mg; vitamin B6 70 mg; vitamin D3 46600 UI; vitamin E 1288 UI; vitamin K3 58 mg; methionine 11.57 g; threonine 35 g; tryptophan 5.83 g; valine 23.33 g; *Bacillus licheniformis* 13.33 × 10⁶ UFC; and *B. subtilis* 13.33 × 10⁶ UFC

capacity). Following, the test ration was pelleted (particle size, 2 to 3 mm).

Fish rearing, maintenance, and water quality parameters

Healthy fish were maintained for 10 days in 250-L fiberglass tanks with continuous aeration under controlled water variables in fresh water. Dissolved oxygen and temperature were evaluated using an oxygen meter (Model Y5512, YSI, Ohio, USA), while the pH was measured using a DMPH-2 pH meter (Digimed, SP, Brazil). Total ammonia levels were determined according to Verdouw et al. (1978), and un-ionized ammonia (NH₃) levels were calculated using a conversion table for fresh water. The water parameters were monitored during the adaptation period (tank, 250 L) and during the experiment in the 16 tank tests.

These water parameters remained stable throughout all acclimation period: temperature (23.1 ± 0.2 °C), pH (6.9 ± 0.4), dissolved oxygen (6.8 ± 0.2 mg/L), total ammonia (0.81 ± 0.03 mg/L), and non-ionized ammonia (0.0041 ± 0.00028 mg/L).

Experimental design

A total of 160 *C. idella* fingerlings (7.33 ± 0.12 g; 8.13 ± 0.12 cm) were maintained in a recirculating aquaculture system and divided into four groups in quadruplicate with ten fish per aquarium (60 L) as follows: the group T0 (basal diet without thymol supplementation), T100 (basal diet supplemented with 100 mg thymol/kg of basal feed), T200 (basal diet supplemented with 200 mg thymol/kg on an as fed basis), and T300 (basal diet supplemented with 300 mg thymol/kg on an as fed basis), following the protocol established by Ran et al. (2016). Fish were fed on one of the tested diets at a feeding rate of 6% of total biomass for 60 consecutive days. The feed protocol was divided into two equal portions and offered to fish twice daily at 09:00 h and 17:00 h.

On day 60 post-feeding (PF), all animals were weighed: body weight, total body length, weight gain, and added body length were calculated for the total period (60 days). Following, the values of specific growth rate (SGR = (final weight – initial weight × 100)/experiment time) were calculated; Feed conversion ratio (FCR = feed intake/weight gain).

Sample collection and tissue preparation

On day 61 PF, two fingerlings from each aquarium ($n = 8$ per treatment) were collected using natural anesthetic (eugenol 150 mg/L) (Ribeiro et al. 2015), followed by spinal cord section according to the ethics committee recommendations. Thereafter, the liver was removed and dissected in a glass dish over ice to evaluate the various parameters.

To evaluate enzymes belonging to the phosphoryl transfer network, liver tissue was washed in SET buffer (0.32 M sucrose, 1 mM EGTA, 10 mM Tris-HCl, pH 7.4) and homogenized (1:10 *w/v*) in the same SET buffer using a Potter-Elvehjem glass homogenizer. The homogenates were centrifuged at $800\times g$ for 10 min at 4 °C. Some of these supernatants were used to evaluate all parameters (except PK activity). To measure hepatic PK activity, after centrifugation, cell pellets were discarded, and the supernatants were once again centrifuged at $10,000\times g$ for 15 min at 4 °C. The supernatants were stored for no more than 2 weeks at -80 °C.

Hepatic antioxidant/oxidant parameters

Hepatic reactive oxygen species (ROS) levels were determined using the DCFH oxidation method described by LeBel et al. (1992) as recently published in detail by Souza et al. (2018) using excitation and emission wavelengths of 485 and 538 nm, respectively. The results were expressed as U DCF per mg of protein. Lipid peroxidation (LOOH) levels were measured as proposed by Montserrat et al. (2003) and reported in detail by Souza et al. (2018). The results were expressed as μmol CHP per g of tissue.

Hepatic SOD activity was evaluated spectrophotometrically as described by Marklund and Marklund (1974), recently published in detail by Souza et al. (2018). Enzymatic activity was expressed as SOD units per mg of protein. Hepatic glutathione S-transferase (GST) activity was measured based on the method described by Habig et al. (1974) and reported in detail by Biazus et al. (2017). Enzymatic activity was expressed as U GST per mg of protein. Hepatic glutathione peroxidase (GPx) activity was measured according to the methodology described by Paglia and Valentine (1967), as reported in detail by Souza et al. (2018). Enzymatic activity was expressed as U GPx per mg of protein. Antioxidant capacity against peroxyl radicals (ACAP) levels was measured according to the protocol

established by Amado et al. (2009) and reported in detail by Souza et al. (2018) using wavelengths of 485 nm (excitation) and 520 nm (emission) for 40 min at 37 °C. The results were expressed as fluorescence units per mg of protein.

Hepatic AK, PK, and LDH activities

Hepatic AK activity was measured with a coupled enzyme assay using hexokinase (HK) and glucose 6-phosphate dehydrogenase (G6PD), according to Dzeja et al. (1999) and published in detail by Baldissera et al. (2019). Hepatic PK activity was assayed according to the protocol established by Leong et al. (1981) and published in detail by Baldissera et al. (2019). The activity was expressed in μmol pyruvate formed per min per mg of protein. The activity was expressed in μmol ATP formed/min/mg of protein. Hepatic lactate dehydrogenase (LDH) activity was measured spectrophotometrically at 340 nm as described by Kaplan et al. (1998) and published in details by Baldissera et al. (2019). The activity was expressed in μmol lactate oxidized per min per mg of protein.

Protein determination

Protein content in liver homogenates was determined by the method of Coomassie blue G dye (Read and Northcote 1981), using serum bovine albumin as the standard.

Statistical analyses

Normality and homoscedasticity were analyzed using the Shapiro–Wilk and Levene tests, respectively. Statistical analysis was performed using bilateral one-way analysis of variance (ANOVA) followed by Bonferroni post-hoc analysis. Results were expressed as mean $\pm$ standard deviation. Statistical manipulations were performed using R-language, v.3.1 (R Development Core Team 2012).

Results

Growth performance

No significant difference was observed between groups regarding length and feed consumption. However, body

Table 2 Performance of grass carp (*Ctenopharyngodon idella*) fed with different levels of thymol in the diet for 60 days

	Starter values		Values at 60 days					FCR ^{&,*}	SGR (%) ^{&,*}
Level	Initial body weight, g	Initial total length, mm	Feed intake, g	Final body weight&, g	Weight gain&, g	Total length, mm	Length gain, mm		
0	7.52	8.00	28.25	8.18 ^b	0.66 ^b	10.25	2.25	42.9 ^a	1.09 ^b
100	7.27	8.16	31.65	10.70 ^a	3.43 ^a	11.78	3.65	9.21 ^b	5.72 ^a
200	7.24	8.225	27.37	8.03 ^b	0.79 ^b	10.58	2.35	34.8 ^a	1.30 ^b
300	7.30	8.20	28.33	8.43 ^b	1.12 ^b	10.58	2.37	25.1 ^a	1.87 ^b
Qualitative statistical analysis									
P-value	0.704	0.714	0.164	0.002	0.006	0.230	0.143	0.001	0.008
CV*	11.46	6.63	9.15	9.63	26.30	9.69	33.88	22.4	16.7

[&] For body weight and weight gain, the difference between groups was illustrated by different letters in the same column (^a and ^b)

SGR, specific growth rate; FCR, feed conversion ratio; CV, coefficient of variation

weight and body weight gain were higher in group T100 compared to all other groups (Table 2). In terms of percentage, the body weight in the fish over 60 days increased 8.7% (T0), 47.1% (T100), 10.9% (T200), and 15.4% (T300). SGR was higher while FCR was lower in fish that consumed 100 mg of thymol/kg compared to other groups (Table 2).

Hepatic antioxidant/oxidant parameters

Hepatic LDH activity was lower in group T100 than in group T0. Hepatic ROS levels were lower in groups T100, T200, and T300 than in group T0, while hepatic LPO levels were lower only in group T100 (Fig. 1). Hepatic SOD activity was higher in group T300 than in

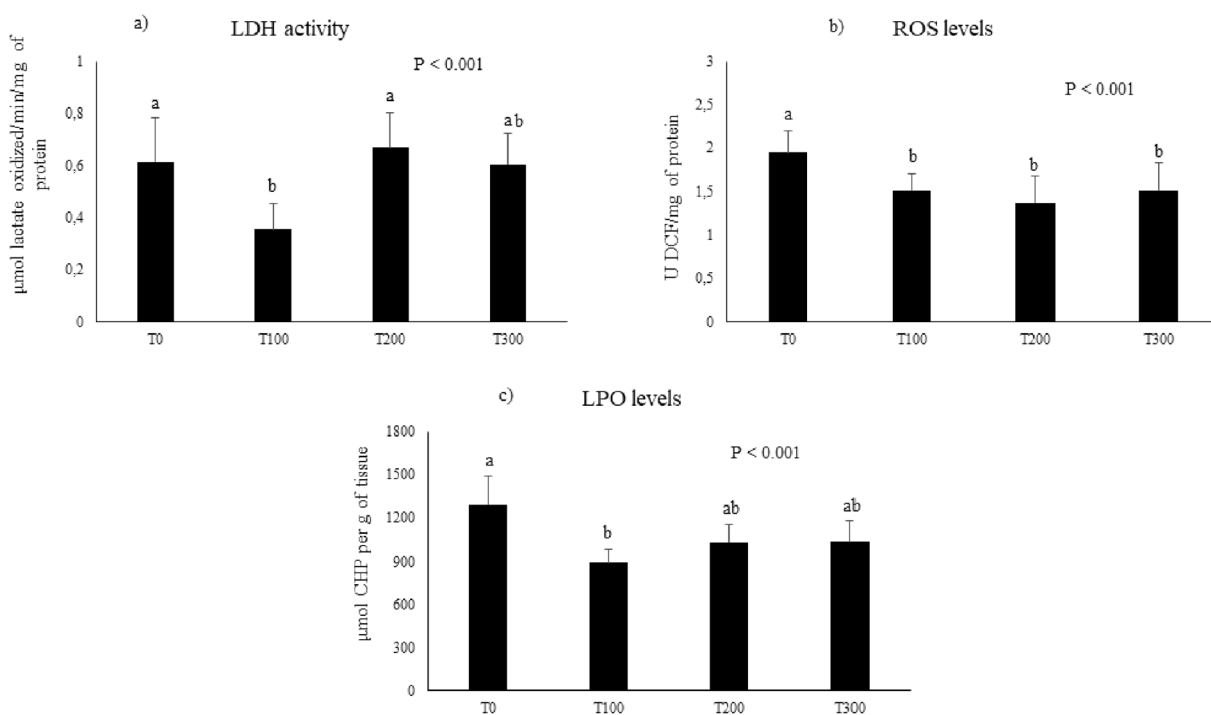


Fig. 1 Hepatic and lactate dehydrogenase (LDH) activities (a), reactive oxygen species (ROS) (b), and lipid peroxidation (LPO) (c) levels of grass carp (*Ctenopharyngodon idella*) fed with

various thymol levels for 60 days. Bars with different letters indicate significant difference between groups at $p < 0.05$

group T0, while hepatic GPx activity was higher in groups T100, T200, and T300 than in group T0. Finally, hepatic ACAP levels were higher in group T100 than in group T0 (Fig. 2).

Hepatic AK, PK, and LDH activities

Hepatic AK activity was higher in group T100 than in group T0. No significant difference was observed between groups with respect to hepatic PK activity (Fig. 3).

Discussion

The performance results of these fish were lower as expected, probably because the fish were raised in non-ideal conditions (water temperature is lower than 24 °C, and these herbivorous fish were fed experimental ration), but it is important to note that the authors do not have a conclusive explanation of the small growth of fish during the experimental period. At 60 days, growth was small in the four groups; however, the increase in

body weight was 47.1% higher in carp fed with 100 mg thymol/kg than in control. Similar results were also observed in other fish species, including Nile tilapia when fed with 1 and 2 mL thymol/kg of feed (Amer et al. 2018). The most important result was that thymol dietary supplementation exerted an ameliorative effect on hepatic antioxidant/oxidant status and energetic metabolism, both of which may be involved in the improvement of grass carp growth performance.

After 60 days, grass carp that received 100 mg thymol/kg of feed showed greater body weight and body weight gain. These results agree with those of Amer et al. (2018), who reported that 1 and 2 mL thymol/kg satisfied all nutritional requirements of Nile tilapia fingerlings and enhanced body weight, body weight gain, and feed intake in all observed periods. Our result corroborates those of Ahmadifar et al. (2011), who found that thymol-carvacrol powder from *O. vulgare* significantly improved growth performance of rainbow trout (*Oncorhynchus mykiss*). Zheng et al. (2009) also reported that a mixture of thymol, carvacrol, and oregano oil improved body weight gain in channel catfish (*Ictalurus*

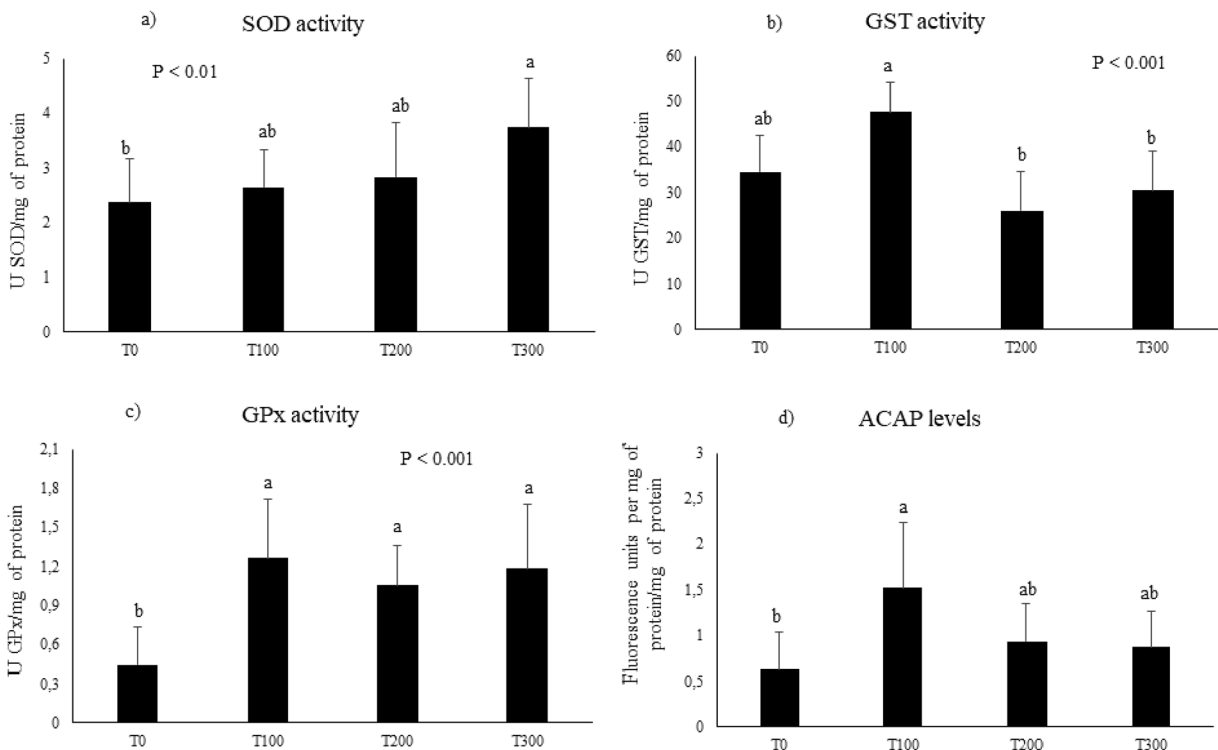
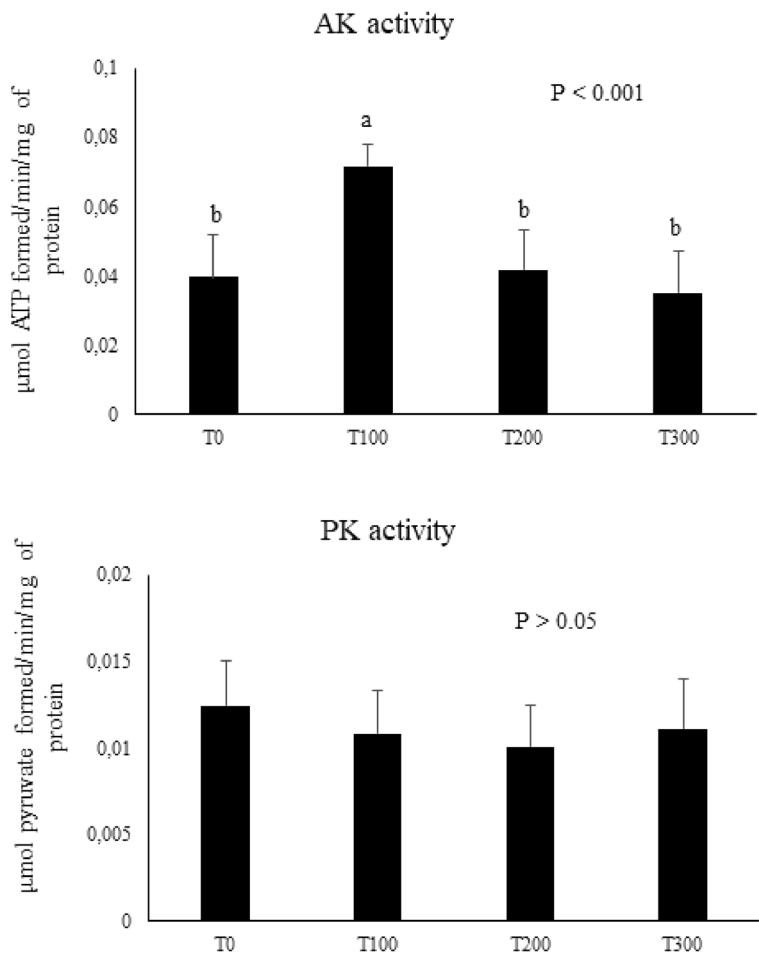


Fig. 2 Hepatic superoxide dismutase (SOD) (a), glutathione S-transferase (GST) (b), glutathione peroxidase (GPx) (c) activities, and antioxidant capacity against peroxyl radicals (ACAP) (d)

levels of grass carp (*Ctenopharyngodon idella*) fed with different thymol levels for 60 days. Bars with different letters indicate significant difference between groups at $p < 0.05$

Fig. 3 Hepatic adenylate kinase (AK) (a), pyruvate kinase (PK) (b) of grass carp (*Ctenopharyngodon idella*) fed with different thymol levels for 60 days. Bars with different letters indicate significant difference between groups at $p < 0.05$



punctatus), suggesting that essential oils containing thymol can improve fish growth performance. In disagreement with our observations, Giannenas et al. (2012) reported that 6 g thymol/kg of feed did not improve body weight or body weight gain of *O. mykiss* over 56 days. Similarly, Anyu et al. (2018) reported that 250 and 500 ppm of thymol for 63 days did not enhance body weight or body weight gain of juvenile Nile tilapia. Despite the fact that several studies have demonstrated the beneficial effects of thymol supplementation on growth performance, the exact pathways involved in this effect remain poorly understood. According to Amer et al. (2018), the enhancement of growth performance in Nile tilapia fingerlings is attributed to enhanced immune and anti-inflammatory responses, as well as to the muscle antioxidant defense system. However, it is important to emphasize that Giannenas et al. (2012) and Anyu et al. (2018) reported better immune responses and antioxidant capacities without

improvement of performance, i.e., these pathways did not contribute to enhance fish performance. Taken together, these findings suggest that 100 mg thymol/kg may be an interesting approach to improve growth performance in grass carp fingerlings.

Some reports have suggested the involvement of improved energy metabolism with increased growth performance, including Baldissera et al. (2019), who studied dietary supplementation with vegetable choline. In the current study, we observed that 100 mg thymol/kg increased hepatic AK activity, suggesting an attempt to double the energetic potential via reconverting two ADP molecules into one ATP and one AMP, contributing to improved intracellular energetic communication and metabolic monitoring in tissues with high and fluctuating energy demands. Although hepatic AK activity did not differ between groups, Baldissera et al. (2019) reported that supplementation with 800 and 1200 mg vegetable choline/kg augmented the activities

of other important enzymes in the phosphoryl transfer network, including hepatic PK, suggesting augmented production of pyruvate and ATP in liver tissue, which can contribute to improved energy transfer and distribution, and to the balance between cellular adenosine triphosphate (ATP) consumption and production. It is important to emphasize that decreased hepatic LDH activity is a consequence of hepatic energetic metabolism improvement elicited by increased AK activity, suggesting that anaerobic metabolism is not necessary to provide energy sufficient to meet the energy requirements of physiological functions in grass carp fingerlings, which can contribute to enhanced fish growth performance and health, as observed by Yousefi et al. (2018). Taken together, the data suggest that hepatic AK activity, as well as the inhibition on hepatic LDH activity may be pathways associated with improved growth performance in grass carp fingerlings supplemented with 100 mg thymol/kg of feed.

To explain the ameliorative effects on AK activity, we measured several oxidative-stress-related parameters, because this enzyme is susceptible to inactivation by free radicals and oxidative damage. As expected, all thymol concentrations reduced hepatic ROS production, while 100 mg thymol/kg of feed reduced lipid peroxidation, in agreement with Amer et al. (2018). According to these authors, 1 and 2 mL thymol/kg of feed decreased lipid peroxidation in the muscle of Nile tilapia, contributing to amelioration of the antioxidant status. It is also important to emphasize that decreased lipid peroxidation can be a direct consequence of reduced ROS levels, because it is the major pro-oxidant free radical linked with lipid peroxidation (Kwiecien et al. 2014). In the present study, thymol enhanced the hepatic antioxidant defense system, as evidenced by the increase in activities of hepatic SOD (400 mg thymol/kg of feed) and GPx (all concentrations), as well by increased ACAP levels (100 mg thymol/kg of feed). In agreement with our observations, Amer et al. (2018) reported that 1 and 2 mL thymol/kg of feed exerted ameliorative effects on antioxidant status via augmentation of muscle glutathione reductase levels and catalase activity, contributing to increased fish growth and health. Similarly, Hoseini and Yousefi (2019) reported that thyme extract (0, 5, 10, and 20 g/kg) elicited a reduction in lipid peroxidation with concomitant increases in SOD and GPx activities in the gut of *O. mykiss* after 14 days of supplementation.

Moreover, it is important to note that the improvement of antioxidant system can be a pathway associated with elevated AK activity, because the ameliorative antioxidant system is required for suitable functioning of the phosphoryl transfer network (Grings et al. 2018). Baldissera et al. (2019) found that augmentation of hepatic PK activity was a consequence of reduction in hepatic ROS and LPO levels that contributed to enhanced growth performance in Nile tilapia fed with 800 and 1200 mg vegetal choline/kg feed. Thus, the improvement in hepatic antioxidant status can be a pathway associated with the improvement on AK activity and consequent increases in grass carp performance.

Based on this evidence, we conclude that 100 mg thymol/kg improve zootechnical performance in grass carp fingerlings under non-ideal conditions. Thymol (100 mg/kg feed) improved hepatic energy metabolism, while practically all thymol-tested concentrations enhanced hepatic antioxidant status, all of which can be pathways involved in the increase of growth performance of grass carp fingerlings. Nevertheless, it is important to note that, according to the statistical analysis, even lower doses of thymol may potentiate carp performance.

Statement: expects data The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Compliance with ethical standards

Ethics committee These experiments were approved by the Animal Welfare Committee of the State University of Santa Catarina (UDESC) under protocol number 8817050418.

Conflict of interest The authors declare that they have no conflict of interest.

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